STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	Dr. Jayaram Boga
Student Name	Srikruthi Varanasi
Roll Number	2520030340
Branch	CSE
Course Outcome	CO1

Case Study Topic: MediFlow Hospital – Appointment Driven Patient Indexing using BST & AVL Trees

Work Done Summary:
Implemented a Plain BST and AVL Tree for MediFlow Hospital patient indexing. Constructed the BST using sorted patient IDs and analysed its height and lookup performance. Built an AVL Tree using the same insertion sequence and applied balancing rotations. Performed deletion operations (30, 70 and 50) and compared AVL performance with Plain BST. Concluded that AVL Trees provide efficient O(log n) lookup performance and maintain balanced height.

Implementation Details:
1. Inserted 13 patient IDs into BST and AVL Tree.
2. Constructed the Plain BST and calculated height.
3. Performed SLA lookup-time analysis.
4. Built AVL Tree and applied RR rotations when required.
5. Deleted Patient IDs 30, 70 and 50.
6. Compared AVL and Plain BST performance.
7. Verified AVL maintained balanced structure after deletions.
8. Uploaded Java implementation to GitHub.

Result: Screenshots / Output
BST Height = 12
AVL Height = 4
BST Lookup Time = 0.0024 ms
AVL Lookup Time = 0.0008 ms
AVL Tree consumed only 0.016% of the 5 ms SLA budget.
Conclusion: AVL Tree provides better search performance and maintains a balanced structure compared to a Plain BST.

GitHub Link:
https://github.com/kruthivx/dsa-project

Student Signature	Faculty Signature